

Figure 1: X: ["Uniform", 0.1,]

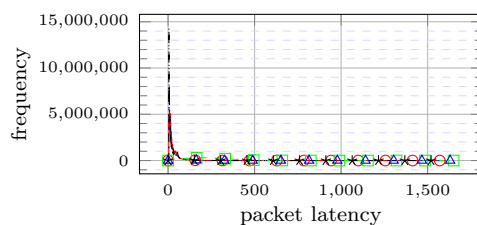
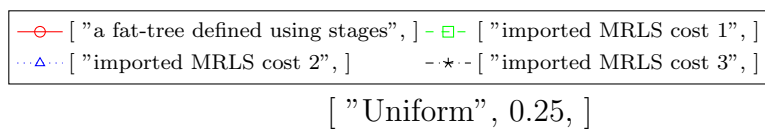


Figure 2: X: ["Uniform", 0.25,]

The following versions used in the simulations.

- heads/alex-dev-8e2a39f34b812569e780c03ed6a7e158aa4b09bc(0.6.3)

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Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 50000,
  statistics_packet_definitions: [
    [ ],
    [ ]],
  general_frequency_divisor: 2,
  topology: topo![
    MultiStage{
      stages: [
        Fat{ bottom_factor: 18, top_factor: 18 },
        Fat{ bottom_factor: 18, top_factor: 18 },
        Fat{ bottom_factor: 18, top_factor: 18 }],
      servers_per_leaf: 18,
      legend_name: "a fat-tree defined using stages"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_5832_18_2916_36" }],
      servers_per_leaf: 18,
      legend_name: "imported MRLS cost 1"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_8748_24_5832_36" }],
      servers_per_leaf: 12,
      legend_name: "imported MRLS cost 2"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_11664_27_8748_36" }],
      servers_per_leaf: 9,
      legend_name: "imported MRLS cost 3"}],
  traffic: ![
    SyntheticTrafficDistribution{
      tasks: 104976,
      load: ![ 0.1, 0.25 ],
      pattern: Uniform,
      message_size: Multimodal{
        sizes: [ 16, 256 ],
        weights: [ 0.9, 0.1 ]},
      legend_name: "Uniform"}],
  maximum_packet_size: 16,
  router: InputOutput{
    virtual_channels: topo![ 3, 4, 4, 4 ],
    virtual_channel_policies: topo![
      OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
        use_neighbour_space: true, aggregate: true },
      OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
        use_neighbour_space: true, aggregate: false },
      LowestLabel,
      EnforceFlowControl,
      Random],
      [
        MapHop{
          hop_to_policy: [
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 3 ]},
            ArgumentVC{
              allowed: [ 3 ]}],
          VecLabel{
            label_vector: [ 0, 64, 80 ]},
            OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
              use_neighbour_space: true, aggregate: true },
            OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
              use_neighbour_space: true, aggregate: false },
            LowestLabel,
            EnforceFlowControl,
            Random],
            [
              MapHop{
                hop_to_policy: [
                  ArgumentVC{
                    allowed: [ 0 ]},
                  ArgumentVC{
                    allowed: [ 0 ]},
                  ArgumentVC{
                    allowed: [ 1 ]},
                  ArgumentVC{
                    allowed: [ 1 ]},
                  ArgumentVC{
                    allowed: [ 2 ]},
                  ArgumentVC{
                    allowed: [ 2 ]},
                  ArgumentVC{
                    allowed: [ 3 ]},
                  ArgumentVC{
                    allowed: [ 3 ]}],
                VecLabel{

```

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    label_vector: [ 0, 64, 80 ]},
    OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
    OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
    LowestLabel,
    EnforceFlowControl,
    Random],
[
  MapHop{
    hop_to_policy: [
      ArgumentVC{
        allowed: [ 0 ]},
      ArgumentVC{
        allowed: [ 0 ]},
      ArgumentVC{
        allowed: [ 1 ]},
      ArgumentVC{
        allowed: [ 1 ]},
      ArgumentVC{
        allowed: [ 2 ]},
      ArgumentVC{
        allowed: [ 2 ]},
      ArgumentVC{
        allowed: [ 3 ]},
      ArgumentVC{
        allowed: [ 3 ]}],
    VecLabel{
      label_vector: [ 0, 64, 80 ]},
      OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
      OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
      LowestLabel,
      EnforceFlowControl,
      Random]],
    allocator: Random,
    buffer_size: 128,
    bubble: false,
    flit_size: 16,
    intransit_priority: false,
    allow_request_busy_port: true,
    output_buffer_size: 64,
    crossbar_frequency_divisor: 1,
    crossbar_delay: 2},
  routing: topo![
    UpDown{ legend_name: "minimal routing" },
    Polarized{ include_labels: true, legend_name: "Polarized original" },
    Polarized{ include_labels: true, legend_name: "Polarized original" },
    Polarized{ include_labels: true, legend_name: "Polarized original" }],
  link_classes: [
    LinkClass{ delay: 2 },
    LinkClass{ delay: 2 },
    LinkClass{ delay: 2 },
    LinkClass{ delay: 2 }],
  launch_configurations: [
    Slurm{ job_pack_size: 1, time: "50-00:00:00" }]}

```